

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)			award (1) each for up to two of following - speeds up a chemical reaction - lowers activation energy - not used up during the reaction doesn't take part in the reaction - neutral answer	2			2		
	(b)	(i)		award (1) for any <u>comparison</u> of active ranges A works in pH range of 0.5-4.5 <u>and</u> B works in pH range 3-8 A works at a lower pH range / B works at a higher pH range A works over a narrower pH range / B works over a wider pH range award (1) for <u>comparison</u> of optimum pH e.g. A works best at pH 2 <u>and</u> B works best at pH 5.5 A works best at a lower pH / B works best at a higher pH award (1) for <u>comparison</u> of activity at given points - both have the same activity at their optimum pH - both have the same activity at pH 3.75 up to maximum (2)						
		(ii)		curve drawn rising from pH 5 then falling to pH 9 (1) peak at pH 7 (1)			2	2		
				Question 11 total	2	0	4	6	0	0